1. Tổng quan dự án

1.1 Mục tiêu

Xây dựng web quiz realtime kiểu Kahoot, gồm 2 vai trò:

Host

- đăng ký / đăng nhập
- tạo quiz
- thêm / sửa / xóa câu hỏi
- tạo phòng chơi
- xem player join
- start game
- chuyển câu tiếp theo
- kết thúc game
- xem leaderboard

Player

- không cần đăng nhập
- nhập PIN + tên để vào phòng
- chờ host start
- nhận câu hỏi realtime
- chọn đáp án
- nhận điểm
- xem leaderboard
- reconnect được khi refresh

2. Kiến trúc hệ thống

2.1 Công nghệ chốt

Backend

- Spring Boot
- Spring Web
- Spring Security
- Spring Data JPA
- MySQL Driver
- Validation
- Lombok
- JWT
- WebSocket + STOMP

Frontend

- React
- Vite
- React Router
- Axios
- SockJS / STOMP client

2.2 Kiến trúc xử lý

REST

Dùng cho:

- auth
- quiz CRUD
- question CRUD
- create room
- join room
- start / next / end
- current question snapshot

- submit answer
- leaderboard snapshot

WebSocket

Dùng cho:

- player joined realtime
- game started realtime
- question changed realtime
- leaderboard updated realtime
- game ended realtime

Nguyên tắc

- **REST = xử lý logic + ghi DB + validate**
- **WebSocket = broadcast event**
- không dùng WebSocket để thay toàn bộ REST

3. Quy ước naming

3.1 JSON

Dùng **camelCase**

Ví dụ:

- currentQuestionIndex
- questionStartedAt
- pointsEarned

3.2 Database

Dùng **snake_case**

Ví dụ:

- game_rooms
- current_question_index

- question_started_at

3.3 API path

Dùng **kebab-case**

Ví dụ:

- /api/game-rooms
- /current-question
- /submit-answer

4. Database schema final

4.1 Bảng users

Column	Type	Constraint
id	BIGINT	PK, auto increment
username	VARCHAR(100)	unique, not null
password_hash	VARCHAR(255)	not null
created_at	DATETIME	not null

Ý nghĩa

Lưu tài khoản host.

4.2 Bảng quizzes

Column	Type	Constraint
id	BIGINT	PK, auto increment
host_id	BIGINT	FK -> users.id, not null
title	VARCHAR(255)	not null
created_at	DATETIME	not null

Quan hệ

- 1 user có nhiều quiz
- 1 quiz thuộc 1 host

4.3 Bảng questions

Column	Type	Constraint
id	BIGINT	PK, auto increment
quiz_id	BIGINT	FK -> quizzes.id, not null
content	TEXT	not null
answer_a	TEXT	not null
answer_b	TEXT	not null
answer_c	TEXT	not null
answer_d	TEXT	not null
correct_answer	VARCHAR(1)	not null
time_limit	INT	not null
created_at	DATETIME	not null

Rule

- correct_answer chỉ nhận: A, B, C, D
- time_limit > 0

4.4 Bảng game_rooms

Column	Type	Constraint
id	BIGINT	PK, auto increment
quiz_id	BIGINT	FK -> quizzes.id, not null
pin	VARCHAR(20)	unique, not null

Column	Type	Constraint
status	VARCHAR(20)	not null
current_question_index	INT	not null, default -1
started_at	DATETIME	null
ended_at	DATETIME	null
question_started_at	DATETIME	null
created_at	DATETIME	not null

Rule

- status chỉ nhận:
    - waiting
    - playing
    - finished
- current_question_index = -1 nghĩa là chưa start

4.5 Bảng players

Column	Type	Constraint
id	BIGINT	PK, auto increment
room_id	BIGINT	FK -> game_rooms.id, not null
name	VARCHAR(100)	not null
score	INT	not null, default 0
session_id	VARCHAR(255)	null
is_connected	BOOLEAN	not null, default true
last_seen_at	DATETIME	null

Column	Type	Constraint
joined_at	DATETIME	not null

Unique

- (room_id, name)

Ý nghĩa

Trong cùng 1 room không cho 2 người trùng tên.

4.6 Bảng player_answers

Column	Type	Constraint
id	BIGINT	PK, auto increment
room_id	BIGINT	FK -> game_rooms.id, not null
player_id	BIGINT	FK -> players.id, not null
question_id	BIGINT	FK -> questions.id, not null
selected_answer	VARCHAR(1)	not null
is_correct	BOOLEAN	not null
points_earned	INT	not null, default 0
answered_at	DATETIME	not null

Unique

- (room_id, player_id, question_id)

Rule

- 1 player chỉ được submit 1 lần cho 1 question trong 1 room

5. Enum và constant chốt cứng

5.1 RoomStatus

- waiting
- playing
- finished

5.2 AnswerOption

- A
- B
- C
- D

5.3 Điểm

- đúng: 100
- sai: 0

6. Business rules final

6.1 Auth

Host

- phải login
- dùng JWT
- được phép:
    - quản lý quiz
    - quản lý question
    - tạo room
    - start / next / end game

Player

- không cần login
- chỉ join bằng:
    - pin

    - name

6.2 Quiz ownership

Mọi API host thao tác với quiz hoặc room phải check:

- quiz đó thuộc host đang login
- room đó được tạo từ quiz của host đang login

Nếu không đúng:

- trả lỗi QUIZ_FORBIDDEN hoặc AUTH_FORBIDDEN

6.3 Start game

Điều kiện

- room đang waiting
- quiz có ít nhất 1 question

Khi start

- status = playing
- currentQuestionIndex = 0
- startedAt = now
- questionStartedAt = now

Event broadcast

- GAME_STARTED
- QUESTION_CHANGED

6.4 Current question

Question hiện tại được xác định bởi:

- currentQuestionIndex

Player được thấy

- questionId
- content
- answerA
- answerB
- answerC
- answerD
- timeLimit
- timeRemaining
- questionNumber
- totalQuestions

Player không được thấy

- correctAnswer

6.5 Timer

Server lưu

- questionStartedAt
- timeLimit

Client tự tính

timeRemaining = timeLimit - (currentTime - questionStartedAt)

Rule

Nếu timeRemaining <= 0

- coi như hết giờ
- không cho submit nữa

Ghi chú

Server không cần push countdown mỗi giây.

6.6 Submit answer

Hợp lệ khi

- room đang playing
- player thuộc room đó
- questionId gửi lên đúng là current question
- player chưa submit câu đó
- chưa hết giờ

Khi submit

- nếu đúng:
    - isCorrect = true
    - pointsEarned = 100
- nếu sai:
    - isCorrect = false
    - pointsEarned = 0

Sau đó:

- lưu vào player_answers
- cộng điểm vào players.score

Event broadcast

- LEADERBOARD_UPDATED

6.7 Next question

Chỉ host được gọi

Logic

- nếu room không ở playing → reject
- nếu còn câu tiếp theo:
    - currentQuestionIndex += 1

    - questionStartedAt = now
    - broadcast QUESTION_CHANGED
- nếu hết câu:
    - status = finished
    - endedAt = now
    - broadcast GAME_ENDED

Không cần chờ tất cả player trả lời

Host có thể bấm next thủ công.

6.8 End game

Chỉ host được gọi

Khi end

- status = finished
- endedAt = now

Event broadcast

- GAME_ENDED

6.9 Reconnect

Player refresh

Frontend phải:

1. đọc localStorage:
    - playerRoomId
    - playerId
    - playerName
2. gọi REST snapshot:
    - room state

    - current question
    - leaderboard
3. reconnect websocket

Backend luôn phải check

- playerId có thuộc roomId không

7. Response format chung

7.1 Success response

```
{
  "success": true,
  "data": {},
  "message": "Success"
}
```

7.2 Error response

```
{
  "success": false,
  "message": "Room not found",
  "errorCode": "ROOM_NOT_FOUND",
  "timestamp": "2026-04-08T21:00:00"
}
```

8. Error code final

8.1 Auth

- AUTH_USERNAME_ALREADY_EXISTS
- AUTH_INVALID_CREDENTIALS
- AUTH_UNAUTHORIZED
- AUTH_FORBIDDEN

8.2 Quiz / Question

- QUIZ_NOT_FOUND

- QUESTION_NOT_FOUND
- QUIZ_FORBIDDEN
- INVALID_QUESTION_DATA

8.3 Room

- ROOM_NOT_FOUND
- ROOM_ALREADY_FINISHED
- ROOM_NOT_PLAYING
- ROOM_ALREADY_STARTED
- ROOM_INVALID_STATUS

8.4 Player

- PLAYER_NAME_ALREADY_EXISTS
- PLAYER_NOT_FOUND
- PLAYER_NOT_IN_ROOM

8.5 Answer / Game

- ANSWER_ALREADY_SUBMITTED
- QUESTION_NOT_CURRENT
- QUESTION_TIME_EXPIRED
- GAME_NOT_STARTED
- GAME_ALREADY_FINISHED

8.6 System

- VALIDATION_ERROR
- INTERNAL_SERVER_ERROR

9. REST API final

9.1 Auth APIs

POST /api/auth/register

Request

```
{
  "username": "teacher1",
  "password": "123456"
}
```

Response

```
{
  "success": true,
  "data": null,
  "message": "Register successful"
}
```

POST /api/auth/login

Request

```
{
  "username": "teacher1",
  "password": "123456"
}
```

Response

```
{
  "success": true,
  "data": {
    "token": "jwt_token",
    "username": "teacher1"
  },
  "message": "Login successful"
}
```

9.2 Quiz APIs

POST /api/quizzes

Auth: host

Request

```
{
  "title": "English Vocabulary Quiz"
}
```

Response

```
{
  "success": true,
  "data": {
    "quizId": 1,
    "title": "English Vocabulary Quiz",
    "hostId": 2
  },
  "message": "Quiz created"
}
```

GET /api/quizzes/my

Auth: host

Response

```
{
  "success": true,
  "data": [
    {
      "quizId": 1,
      "title": "English Vocabulary Quiz",
      "createdAt": "2026-04-08T20:00:00"
    }
  ],
  "message": "Quiz list fetched"
}
```

GET /api/quizzes/{quizId}

Auth: host

Response


```
{
  "success": true,
  "data": {
    "quizId": 1,
    "title": "English Vocabulary Quiz",
    "hostId": 2,
    "createdAt": "2026-04-08T20:00:00",
    "questions": [
      {
        "questionId": 101,
        "content": "What is the synonym of happy?",
        "answerA": "Sad",
        "answerB": "Joyful",
        "answerC": "Angry",
        "answerD": "Tired",
        "correctAnswer": "B",
        "timeLimit": 20
      }
    ]
  },
  "message": "Quiz detail fetched"
}
```


PUT /api/quizzes/{quizId}

Auth: host

Request


```
{
  "title": "Updated quiz title"
}
```


Response


```
{
  "success": true,
  "data": null,
```

```
  "message": "Quiz updated"
}
```


DELETE /api/quizzes/{quizId}

Auth: host

Response


```
{
  "success": true,
  "data": null,
  "message": "Quiz deleted"
}
```


9.3 Question APIs

POST /api/quizzes/{quizId}/questions

Auth: host

Request


```
{
  "content": "What is the synonym of happy?",
  "answerA": "Sad",
  "answerB": "Joyful",
  "answerC": "Angry",
  "answerD": "Tired",
  "correctAnswer": "B",
  "timeLimit": 20
}
```


Response


```
{
  "success": true,
  "data": {
    "questionId": 101
  },
```

```
  "message": "Question added"
}
```


PUT /api/questions/{questionId}

Auth: host

Request


```
{
  "content": "Updated question",
  "answerA": "A1",
  "answerB": "B1",
  "answerC": "C1",
  "answerD": "D1",
  "correctAnswer": "A",
  "timeLimit": 15
}
```


Response


```
{
  "success": true,
  "data": null,
  "message": "Question updated"
}
```


DELETE /api/questions/{questionId}

Auth: host

Response


```
{
  "success": true,
  "data": null,
  "message": "Question deleted"
}
```

9.4 Game room APIs

POST /api/game-rooms

Auth: host

Request

```
{
  "quizId": 1
}
```

Response

```
{
  "success": true,
  "data": {
    "roomId": 10,
    "quizId": 1,
    "pin": "123456",
    "status": "waiting",
    "currentQuestionIndex": -1,
    "createdAt": "2026-04-08T20:10:00"
  },
  "message": "Room created"
}
```

POST /api/game-rooms/join

Auth: none

Request

```
{
  "pin": "123456",
  "name": "Phuong"
}
```

Response

```
{
  "success": true,
  "data": {
```

```
    "roomId": 10,
    "playerId": 5,
    "name": "Phuong",
    "score": 0
  },
  "message": "Joined successfully"
}
```


GET /api/game-rooms/{roomId}

Auth: none

Response


```
{
  "success": true,
  "data": {
    "roomId": 10,
    "quizId": 1,
    "pin": "123456",
    "status": "playing",
    "currentQuestionIndex": 0,
    "totalQuestions": 5,
    "playerCount": 12
  },
  "message": "Room state fetched"
}
```


GET /api/game-rooms/{roomId}/players

Auth: none

Response


```
{
  "success": true,
  "data": [
    {
      "playerId": 5,
```

```
    "name": "Phuong",
    "score": 100,
    "joinedAt": "2026-04-08T20:11:00"
  }
 ],
 "message": "Players fetched"
}
```


POST /api/game-rooms/{roomId}/start

Auth: host

Response


```
{
 "success": true,
 "data": null,
 "message": "Game started"
}
```


POST /api/game-rooms/{roomId}/next

Auth: host

Response


```
{
 "success": true,
 "data": null,
 "message": "Next question"
}
```


POST /api/game-rooms/{roomId}/end

Auth: host

Response


```
{
 "success": true,
```

```
  "data": null,
  "message": "Game ended"
}
```


9.5 Game APIs

GET /api/game-rooms/{roomId}/current-question

Auth: none

Response


```
{
  "success": true,
  "data": {
    "questionId": 101,
    "content": "What is the synonym of happy?",
    "answerA": "Sad",
    "answerB": "Joyful",
    "answerC": "Angry",
    "answerD": "Tired",
    "timeLimit": 20,
    "timeRemaining": 14,
    "questionNumber": 1,
    "totalQuestions": 5
  },
  "message": "Current question fetched"
}
```


POST /api/game-rooms/{roomId}/submit-answer

Auth: none

Request


```
{
  "playerId": 5,
  "questionId": 101,
  "selectedAnswer": "B"
}
```

Response


```
{
  "success": true,
  "data": {
    "isCorrect": true,
    "pointsEarned": 100,
    "totalScore": 100
  },
  "message": "Answer submitted"
}
```


GET /api/game-rooms/{roomId}/leaderboard

Auth: none

Response


```
{
  "success": true,
  "data": [
    {
      "playerId": 5,
      "name": "Phuong",
      "score": 300
    },
    {
      "playerId": 6,
      "name": "An",
      "score": 200
    }
  ],
  "message": "Leaderboard fetched"
}
```


GET /api/game-rooms/{roomId}/players/{playerId}/answers

Auth: none

Response


```
{
  "success": true,
  "data": [
    {
      "questionId": 101,
      "selectedAnswer": "B",
      "isCorrect": true,
      "pointsEarned": 100,
      "answeredAt": "2026-04-08T20:15:00"
    }
  ],
  "message": "Player answers fetched"
}
```


10. API cần token

10.1 Cần token

- POST /api/quizzes
- GET /api/quizzes/my
- GET /api/quizzes/{quizId}
- PUT /api/quizzes/{quizId}
- DELETE /api/quizzes/{quizId}
- POST /api/quizzes/{quizId}/questions
- PUT /api/questions/{questionId}
- DELETE /api/questions/{questionId}
- POST /api/game-rooms
- POST /api/game-rooms/{roomId}/start
- POST /api/game-rooms/{roomId}/next
- POST /api/game-rooms/{roomId}/end

10.2 Không cần token

- POST /api/game-rooms/join
- GET /api/game-rooms/{roomId}
- GET /api/game-rooms/{roomId}/players
- GET /api/game-rooms/{roomId}/current-question
- POST /api/game-rooms/{roomId}/submit-answer
- GET /api/game-rooms/{roomId}/leaderboard
- GET /api/game-rooms/{roomId}/players/{playerId}/answers

11. WebSocket spec final

11.1 Endpoint connect

/ws

11.2 Prefix

- client send: /app
- server push: /topic

11.3 Subscribe topics

- /topic/rooms/{roomId}
- /topic/rooms/{roomId}/players
- /topic/rooms/{roomId}/question
- /topic/rooms/{roomId}/leaderboard

12. Event format final

Format chung

```
{
  "type": "EVENT_NAME",
  "roomId": 10,
  "payload": {}
}
```

Event type chỉ dùng

- PLAYER_JOINED
- GAME_STARTED
- QUESTION_CHANGED
- LEADERBOARD_UPDATED
- GAME_ENDED

13. Event payload samples

13.1 PLAYER_JOINED

Topic: /topic/rooms/{roomId}/players

```
{
  "type": "PLAYER_JOINED",
  "roomId": 10,
  "payload": {
    "playerId": 5,
    "name": "Phuong"
  }
}
```

13.2 GAME_STARTED

Topic: /topic/rooms/{roomId}

```
{
  "type": "GAME_STARTED",
  "roomId": 10,
  "payload": {
    "currentQuestionIndex": 0,
    "questionStartedAt": "2026-04-08T20:20:00",
    "serverTime": "2026-04-08T20:20:00"
  }
}
```

13.3 QUESTION_CHANGED

Topic: /topic/rooms/{roomId}/question


```
{
  "type": "QUESTION_CHANGED",
  "roomId": 10,
  "payload": {
    "questionId": 101,
    "content": "What is the synonym of happy?",
    "answerA": "Sad",
    "answerB": "Joyful",
    "answerC": "Angry",
    "answerD": "Tired",
    "timeLimit": 20,
    "questionNumber": 1,
    "totalQuestions": 5,
    "questionStartedAt": "2026-04-08T20:20:00",
    "serverTime": "2026-04-08T20:20:00"
  }
}
```


13.4 LEADERBOARD_UPDATED

Topic: /topic/rooms/{roomId}/leaderboard


```
{
  "type": "LEADERBOARD_UPDATED",
  "roomId": 10,
  "payload": [
    {
      "playerId": 5,
      "name": "Phuong",
      "score": 100
    },
    {
      "playerId": 6,
      "name": "An",
      "score": 0
```

```
    }
  ]
}
```

13.5 GAME_ENDED

Topic: /topic/rooms/{roomId}

```
{
  "type": "GAME_ENDED",
  "roomId": 10,
  "payload": {
    "endedAt": "2026-04-08T20:30:00"
  }
}
```

14. Realtime flow final

14.1 Player join

1. Player gọi REST POST /join
2. Server tạo player record
3. Server broadcast PLAYER_JOINED

14.2 Host start

1. Host gọi REST POST /start
2. Server update DB
3. Server broadcast:
    - GAME_STARTED
    - QUESTION_CHANGED

14.3 Player submit

1. Player gọi REST POST /submit-answer
2. Server validate + lưu DB + cộng điểm

3. Server broadcast LEADERBOARD_UPDATED

14.4 Host next

1. Host gọi REST POST /next
2. Server update current question
3. Server broadcast QUESTION_CHANGED
4. Nếu hết câu thì broadcast GAME_ENDED

14.5 Host end

1. Host gọi REST POST /end
2. Server update status
3. Server broadcast GAME_ENDED

15. Frontend routes final

Host

- /login
- /register
- /host/quizzes
- /host/quizzes/:quizId
- /host/rooms/:roomId

Player

- /play
- /play/room/:roomId

16. localStorage keys final

Host

- accessToken

Player

- playerRoomId
- playerId
- playerName

17. Frontend behavior final

17.1 Host side

Login/Register

- login thành công → lưu accessToken
- dùng axios interceptor gắn Bearer token

Quiz management

- list quiz
- create quiz
- view quiz detail
- add/edit/delete question

Room page

- hiển thị PIN
- hiển thị player list realtime
- start / next / end
- hiển thị câu hiện tại
- hiển thị leaderboard realtime

17.2 Player side

Join

- nhập pin
- nhập name
- join thành công → lưu localStorage

Lobby

- chờ host start
- subscribe websocket để nhận event

Game screen

- nhận question realtime
- hiện countdown
- submit 1 lần
- disable sau khi submit hoặc hết giờ

Leaderboard

- cập nhật realtime

Reconnect

- đọc localStorage
- gọi snapshot APIs
- resubscribe socket

18. Package structure backend

```
com.example.quizweb
├── config
│   ├── SecurityConfig
│   ├── JwtAuthenticationFilter
│   ├── JwtTokenProvider
│   ├── WebSocketConfig
│   └── CorsConfig
├── controller
│   ├── AuthController
│   ├── QuizController
│   ├── QuestionController
```

```
│   ├── GameRoomController
│   └── GameController
├── websocket
│   ├── GameEventPublisher
│   ├── WebSocketEventListener
│   └── dto
├── entity
│   ├── User
│   ├── Quiz
│   ├── Question
│   ├── GameRoom
│   ├── Player
│   └── PlayerAnswer
├── repository
│   ├── UserRepository
│   ├── QuizRepository
│   ├── QuestionRepository
│   ├── GameRoomRepository
│   ├── PlayerRepository
│   └── PlayerAnswerRepository
├── service
│   ├── AuthService
│   ├── QuizService
│   ├── QuestionService
│   ├── GameRoomService
│   ├── GameService
│   └── LeaderboardService
├── dto
```

```
│   ├── request
│   └── response
├── exception
│   ├── ErrorCode
│   ├── ApiException
│   └── GlobalExceptionHandler
└── common
    └── ApiResponse
```

19. Security final

19.1 JWT áp dụng cho host

Protected endpoints:

- quiz APIs
- question APIs
- create room
- start / next / end

19.2 Public endpoints

- join room
- room state
- current question
- submit answer
- leaderboard

19.3 Lưu ý

Dù player không có JWT, backend vẫn phải check:

- playerId thuộc roomId

20. Validation final

20.1 Auth

- username không rỗng
- password không rỗng

20.2 Quiz

- title không rỗng

20.3 Question

- content không rỗng
- 4 đáp án không rỗng
- correctAnswer phải là A/B/C/D
- timeLimit > 0

20.4 Join room

- pin không rỗng
- name không rỗng

20.5 Submit answer

- selectedAnswer phải là A/B/C/D

21. Git workflow final

Branch

- main
- develop

Working branches

- be-auth-quiz
- be-game-websocket
- fe-host
- fe-player

Rule

- không push thẳng main
- merge vào develop trước
- test full flow ổn mới merge main

22. Thứ tự tích hợp hệ thống

Test theo đúng flow này:

1. host register
2. host login
3. host create quiz
4. host add questions
5. host create room
6. player join room
7. host thấy player realtime
8. host start game
9. player nhận question realtime
10. player submit answer
11. leaderboard update realtime
12. host next question
13. hết câu hoặc host end game
14. player thấy game ended

23. Phân công final

BE1: Auth + quiz

- setup project
- config DB

- config security/JWT
- tạo toàn bộ entity
- auth
- quiz CRUD
- question CRUD
- common response
- exception handling

BE2: Game + websocket

- room APIs
- game logic
- submit answer
- leaderboard
- websocket config
- event broadcast

FE1: Host

- host auth UI
- host quiz management UI
- host room UI

FE2:Player

- player join/play UI
- timer
- leaderboard
- websocket subscribe
- reconnect flow